

NAKSHA

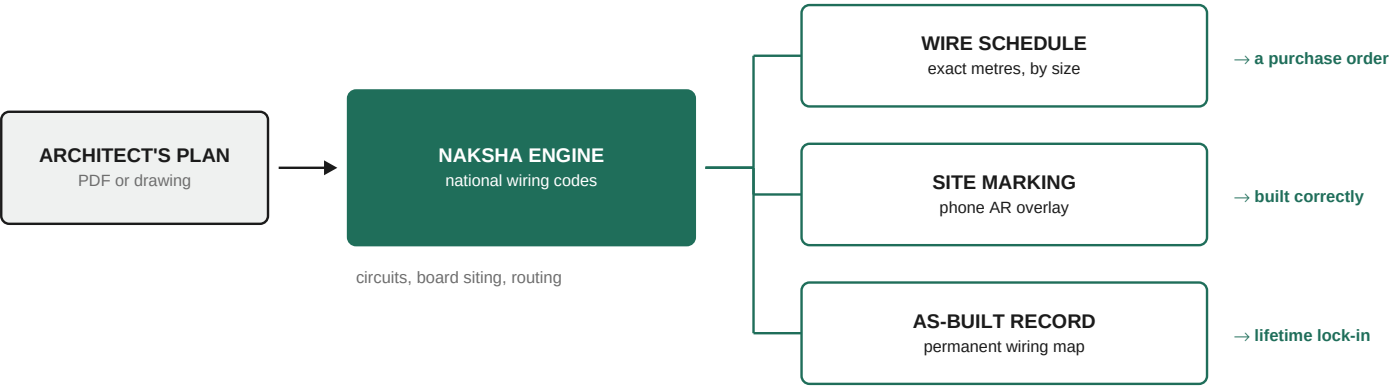
V-Guard designs the wiring, then sells the wire

Track	Track 4: Reimagining V-Guard for an AI Powered Era
Team name	
Institute	
Team members	

1. Brief of the Idea / Solution

Wire is the most commoditised product V-Guard sells. It competes on price, brand recall, and whatever the electrician happens to be carrying. **NAKSHA changes what is being sold: not a coil of wire, but a designed electrical installation that specifies V-Guard wire.**

An owner, contractor or electrician uploads the architect's floor plan and answers a few questions about appliances. NAKSHA returns a complete residential electrical design: circuit split, distribution board siting, conduit routing and point placement, generated against the national wiring and earthing codes.



One upload, three outputs. The first is effectively a purchase order, the second gets the job built correctly, and the third never leaves the house.

Indian homes are wired from memory. The electrician works without a drawing, improvises conduit routes and leaves no record behind. Roughly 42% of building fires in India are attributed to electrical short circuits, a phrase that conceals ageing wiring, overloaded circuits and substandard material. The codes already exist and regulations already require a licensed contractor. **Nothing translates those rules into a drawing anyone on an Indian residential site will actually use, and with no drawing there is no record.**

2. Novelty and Innovation

The components are not new, and the proposal is stronger for saying so. Automated electrical takeoff is a mature commercial category: tools such as drawer.ai already accept PDF drawings and return quantities and wire lengths with marked-up routing, and Kreo, Trimble Accubid and PlanSwift serve the same market. Robotic site layout is also solved, with Dusty Robotics having printed coordinated models onto more than 300 million square feet of slab.

All of it serves Western commercial contractors bidding large projects. Our contribution is positional:

- **Indian codes as the rule set**, which none of the existing tools encode.
- **Residential scale**, where no drawing exists today, rather than commercial scale where one already does.
- **Distribution through a channel V-Guard already owns**. Every major Indian wire brand has already digitised its electrician relationship. Polycab's Experts platform alone reaches around 2.5 lakh electricians and retailers. **Every one of them is a loyalty scheme: scan a code, collect points, redeem a gift. Nobody has put a tool in that channel that does any actual work.**
- **The as-built record**, which no takeoff tool produces because its users leave the site and never return.

3. Technology Proposed

- **Plan understanding.** Vectorise the uploaded plan, then segment rooms, walls and openings. Room type is inferred from dimensions and adjacency, with human confirmation rather than blind trust.
- **Rule engine.** The national wiring and earthing codes encoded as explicit, auditable constraints: separate lighting and power circuits, dedicated ways for heavy loads, points per circuit, earthing requirements and diversity factors. Deterministic and inspectable, not a language model guessing at safety rules.
- **Layout optimisation.** The core computational problem. Place the distribution board and route conduits to minimise total wire length and voltage drop, subject to the rule set and to buildable chase geometry. A graph problem over the room adjacency network, solved with shortest path routing plus a search over board placement.
- **Bill of quantities.** Wire length per size with waste allowance, plus conduit, boxes and switchgear, mapped directly onto V-Guard SKUs.
- **Site delivery.** A phone based AR overlay puts the design on the actual wall. Cheap, works on any modern handset, needs no equipment on site. A trained marking partner is offered in metros, and a robotic layout flagship exists as a showcase rather than as a margin line.
- **As-built capture.** The electrician confirms each run as it is laid, before plastering. The corrected model becomes the homeowner's permanent record and the training signal that improves the generator.

4. End Consumer and Business Value

For the consumer, a home wired to code rather than to habit, an itemised material list that makes overcharging visible, and a permanent map of where every cable runs, which is what makes future renovation and fault finding safe instead of speculative.

For the electrician, the load calculation done, wire sizes decided, the material list totalled, and a defensible answer when the owner asks why it costs what it costs. He does not want loyalty points.

For V-Guard, four things:

- **Every design is a purchase order,** because the output is a wire schedule in metres by size. Specifying the material is a stronger position than advertising to whoever walks into a shop. This is the Asian Paints move, which used Beautiful Homes services to escape commodity paint competition, and which Berger followed with Express Painting.
- **It captures the decision years earlier.** Wiring is specified during construction, before the owner has any brand opinion, and it is never re-shopped. You cannot rewire a finished house.
- **It opens a portfolio funnel.** The same plan positions the water heater, chimney, RO and gas point, all V-Guard categories, all decided at construction stage.
- **It is a margin story.** FY25-26 revenue grew 7.0% to Rs 5,966 crore while PAT fell 1.7% to Rs 308 crore. Commodity wire cannot fix that; specified systems can.

Risks we acknowledge

Indian residential plans are often scanned, hand marked or dimensionally unreliable, so the first release must degrade gracefully to guided manual tracing. Because a licensed contractor is legally responsible for the installation, the tool must be positioned as a decision aid he signs off, not as designer of record. AR marking accuracy on a live site is unproven at the tolerance conduit chasing needs, so we promise dimensioned guidance rather than millimetre placement. **The real risk is adoption, not technology:** the tool must visibly save an electrician time on load calculation and estimating on day one, or it will be ignored. That is what we would test first.